

Which conduction parameters can an electrocardiogram determine?

A calibrated identifiability characterization of the His-Purkinje system

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Abstract: Conduction models are routinely fit to the electrocardiogram (ECG) and their parameter values reported, but the prior question of which conduction parameters an ECG can determine at all is rarely asked, and a model can fit while its reported number carries no information. We characterize the identifiability of seven His-Purkinje and myocardial conduction parameters from the 12-lead ECG at fixed anatomy, using a single amortized neural posterior estimator with a formal calibration audit (simulation-based calibration, expected coverage, and TARP). Against an explicit observation-noise floor (waveform noise, white Gaussian $\sigma = 0.025$ mV per sample per lead, applied before feature extraction), the prior-averaged posterior contraction orders the parameters as interventricular delay (about 0.15) and myocardial velocity (about 0.35) well constrained, RV initial Purkinje extent moderately constrained (about 0.63), and LV extent, branch angle, and branch repulsivity diffuse (about 1.0 to 1.2). Per-parameter conformal recalibration moves the joint posterior from overconfident (TARP ATC -0.057) to essentially calibrated ($+0.007$) without disturbing that ordering. The forward is a pseudo-ECG in an unbounded homogeneous volume conductor with arbitrary-unit amplitudes, not a torso model; there is no measured ECG in the study, and the noise model is white where real ECG noise is not, so the waveform bound is optimistic in a stated direction of unestablished magnitude. Identifiability statements are conditional on that floor and that forward.

Index Terms: Simulation-based inference, neural posterior estimation, identifiability, calibration, His-Purkinje system, electrocardiogram, cardiac digital twin.

I. INTRODUCTION

The His-Purkinje conduction system sets the sequence in which the two ventricles activate: an insulated network of fast-conducting fibers carries the activation wavefront from the atrioventricular node to a distributed set of terminal junctions on the endocardium, and the resulting breakthrough pattern shapes the whole of ventricular depolarization (1). This network cannot be observed directly in a living patient without an invasive study, so the routine noninvasive window onto ventricular activation is the surface ECG, a far-field, spatially integrating measurement (2). The clinical stakes are concrete: interventricular electrical delay is a measurable predictor of who benefits from cardiac resynchronization therapy (3,4). A conduction pa-

rameter recoverable from the ECG would therefore be one that could inform therapy.

This has motivated a line of work that fits a conduction model to a measured or simulated ECG and reports estimated parameter values. Grandits et al. build a cardiac digital twin from the surface ECG and infer properties of the ventricular conduction system (5); a related line learns a fractal Purkinje network from the ECG with Bayesian optimization plus approximate Bayesian computation (6). Both return conduction parameters. Neither first asks the question that logically precedes any reported value: given this ECG and this forward model, which conduction parameters can the ECG determine at all, and how well.

That question is not rhetorical, because the ECG-to-conduction map is non-injective. Grandits et al. demonstrate that distinct activation maps can generate identical surface ECGs (5), which means a fitting procedure can converge, report a tight-looking value, and have that value be a consequence of the optimizer or the prior rather than a quantity the data constrain. A model can fit and its reported number still carry no information. Reporting a point estimate, or an ensemble spread, without a calibration audit does not distinguish an identified parameter from an unconstrained one.

This work addresses that gap. We deliver a quantified identifiability characterization of seven conduction parameters at fixed anatomy. The primary quantity is a per-parameter *contraction*, defined as the posterior standard deviation divided by the prior standard deviation and reported against the waveform noise floor under which the estimator is trained. The characterization is produced by a single amortized neural posterior estimator (NPE), a normalizing flow trained once over the parameter space at fixed geometry (7,8), and its reliability is then tested with formal calibration diagnostics. The contribution is a measurement of the identifiability structure, not a new inference method and not a claim about how closely the forward reproduces any measurement.

The calibration step is load-bearing precisely because it is easy to over-read. A calibration diagnostic checks whether the posterior’s stated uncertainty is self-consistent; it is necessary before any contraction is interpreted, but it is not itself evidence that the ECG carries information about a parameter, because a posterior equal to the prior is

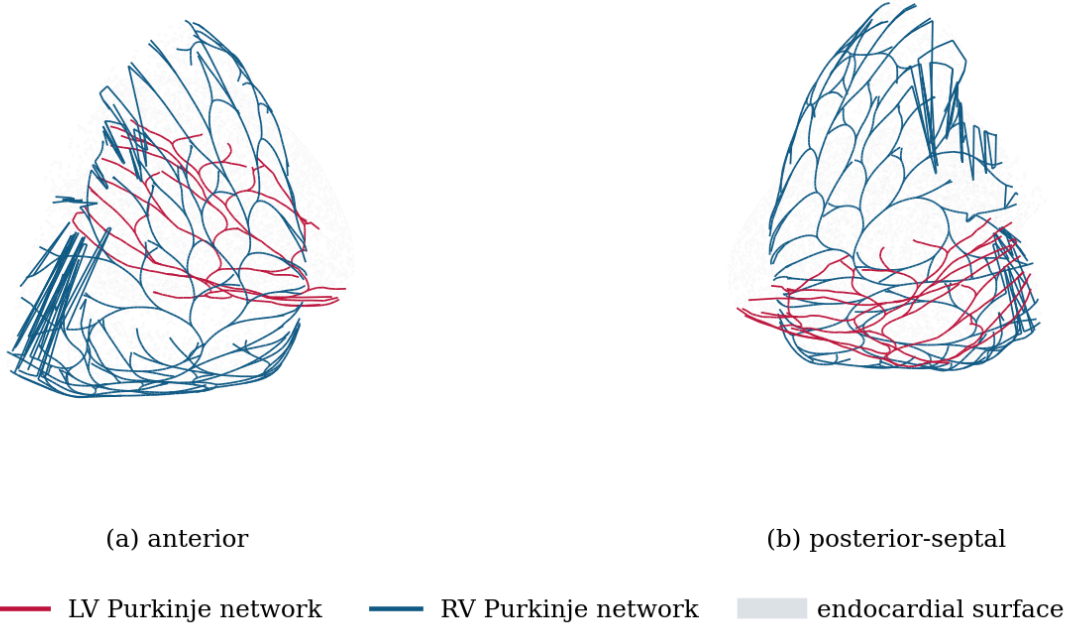


Figure 1: The crt demo endocardial geometry (light point cloud) with a representative grown fractal Purkinje network, LV in red and RV in blue, from two viewpoints. The identifiability characterization fixes this anatomy and infers only the conduction parameters.

perfectly calibrated and maximally diffuse (9,10,11). We therefore report contraction against the stated floor and audit calibration separately, and we read a near-prior posterior as a weakly constrained parameter under that floor, not as an estimator failure.

II. METHODS

II.A. Forward model and simulator

The forward map is deterministic given the conduction parameters θ . A fractal Purkinje tree is grown on the endocardium and its activation computed with a fast iterative eikonal method; the resulting activation-time field drives a 12-lead pseudo-ECG computed as an unbounded homogeneous volume-conductor recovery of the extracellular potential (2). Working at the activation-time level rather than a full ionic solve is what makes a large parameter sweep tractable, and monodomain-to-surface recovery is defensible because the monodomain and bidomain surface potentials differ very little (12,13). The full pipeline is shown in Figure 2. The fixed endocardial geometry and a representative grown fractal Purkinje network are shown in Figure 1.

Two properties of this forward matter for the interpretation. First, it is deterministic in θ ; there is no simulator noise, so an explicit observation noise model is mandatory or calibration is meaningless. Second, the unbounded homogeneous recovery has no absolute amplitude calibration and is known to under-estimate depolarization amplitude by over an order of magnitude relative to a bounded torso forward, a deficit that is distance-dependent and therefore

acts as a differential distortion across leads rather than a single uniform gain (14). Amplitudes are reported in arbitrary units scaled to a stated mV operating point.

II.B. Parameters, priors, and noise floor

The inference target is the seven-dimensional vector θ with components `cv` (conduction velocity), `delta_iv` (LV-RV interventricular delay), `init_length_lv` and `init_length_rv` (LV and RV initial Purkinje extent), `branch_angle`, `w` (branch repulsivity), and `cv_myo` (myocardial conduction velocity). The prior is an independent uniform box over physiological ranges. Parameter order is canonical and fixes the column order of every posterior artifact.

Because the forward is deterministic, the observation noise is stated explicitly. The contraction spectrum is measured under a waveform noise floor: white Gaussian $\sigma = 0.025$ mV per sample per lead, added to the 12-lead ECG before feature extraction. A separate feature-channel model (amplitude $\sigma = 0.05$ mV, timing $\sigma = 5$ ms) is used only as the reference noise for the feature Cramer-Rao bound in the sufficiency analysis (Section IV-C); it is not the floor under which the spectrum is measured. The noise is white where real ECG noise is colored, so the resulting bound is optimistic in a stated direction.

II.C. Inference and calibration

A single normalizing-flow NPE is trained once over θ at fixed geometry (`cardiac_demo`) from a seeded sweep of 5000 simulations that stores, per sample, both the engineered feature vector and the full 12-lead waveform, so the paired feature-versus-waveform comparison costs training time, not simulation time. The primary quantity is the per-

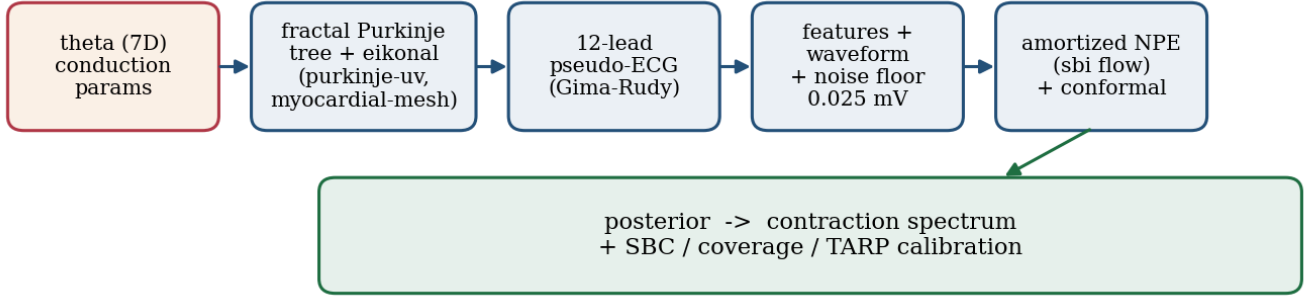


Figure 2: The identifiability pipeline. Conduction parameters θ generate a fractal Purkinje tree and eikonal activation, then a 12-lead pseudo-ECG; engineered features and the full waveform, each under an explicit noise floor, are the observation for a single amortized NPE whose posterior is summarized as a contraction spectrum and audited by SBC, expected coverage, and TARP.

parameter contraction, posterior standard deviation over prior standard deviation; values near or above one indicate a posterior no tighter than the prior.

Calibration is a three-stage stack. Simulation-based calibration (SBC) tests whether posterior rank statistics of prior draws are uniform (9). Per-parameter conformal recalibration is then applied to the raw flow output. Expected coverage and the TARP test assess whether credible regions attain their nominal frequentist coverage (11), motivated by the documented failure mode in which NPE posteriors can be overconfident (10). In the sbi sign convention a negative TARP ATC denotes an underdispersed (overconfident) posterior and a value near zero denotes calibration. Simulation-budget sensitivity is assessed by recomputing contraction as the training budget is varied (15).

III. RESULTS

III.A. The calibrated identifiability spectrum

Against the waveform noise floor the per-parameter contraction orders cleanly (Figure 3): interventricular delay (`delta_iv`, about 0.15) and myocardial velocity (`cv_myo`, about 0.35) are well constrained; RV initial Purkinje extent (`init_length_rv`, about 0.63) and global conduction velocity (`cv`, about 0.67) are moderately constrained; and LV initial extent (`init_length_lv`), branch repulsivity (`w`), and branch angle (`branch_angle`) are diffuse, at about 1.0 to 1.2. Contraction at or above one for the diffuse block is not a defect: it is honest calibration revealing that the posterior for those parameters is no tighter than the prior.

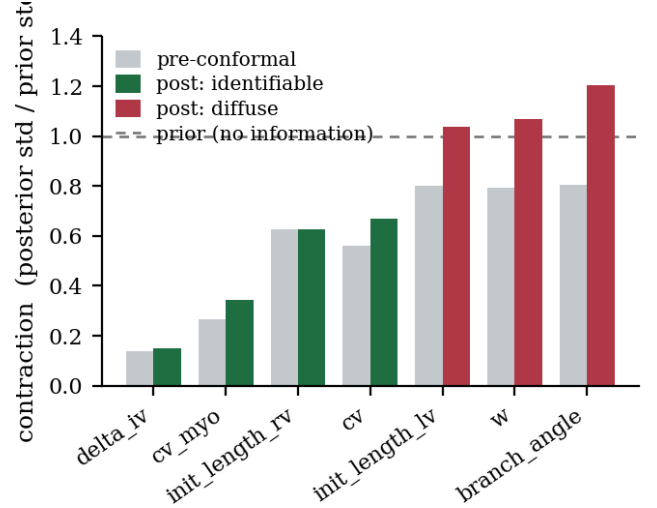


Figure 3: Calibrated identifiability spectrum. Per-parameter contraction (posterior std / prior std) before and after conformal recalibration, sorted by the post-conformal value. Green marks identifiable parameters, red the diffuse block; the dashed line at one is the prior (no information). Measured against the waveform noise floor, $\sigma = 0.025$ mV per sample per lead.

The ordering has a mechanistic reading. Interventricular delay drives interlead timing and myocardial velocity drives QRS duration, both robust high-signal features; RV extent rides an early, comparatively isolated precordial feature (early V1-V2 forces from RV apical breakthrough); the diffuse trio barely moves the QRS and is interaction-dominated, matching an independent forward Sobol analysis (16). The full joint posterior (Figure 4) makes the same point directly: the marginals for interventricular delay and myocardial velocity are tight around the ground truth, while branch angle, branch repulsivity, and LV extent stay broad, and the mild `delta_iv` to `cv_myo` tilt is a ridge, not a degeneracy.

III.B. Calibration

The calibration audit is shown in Figure 5. Before recalibration the empirical coverage sits below nominal (overconfident); after per-parameter conformal recalibration it tracks the diagonal, and the joint TARP ATC moves from -0.057 (overconfident) to $+0.007$ (essentially calibrated,

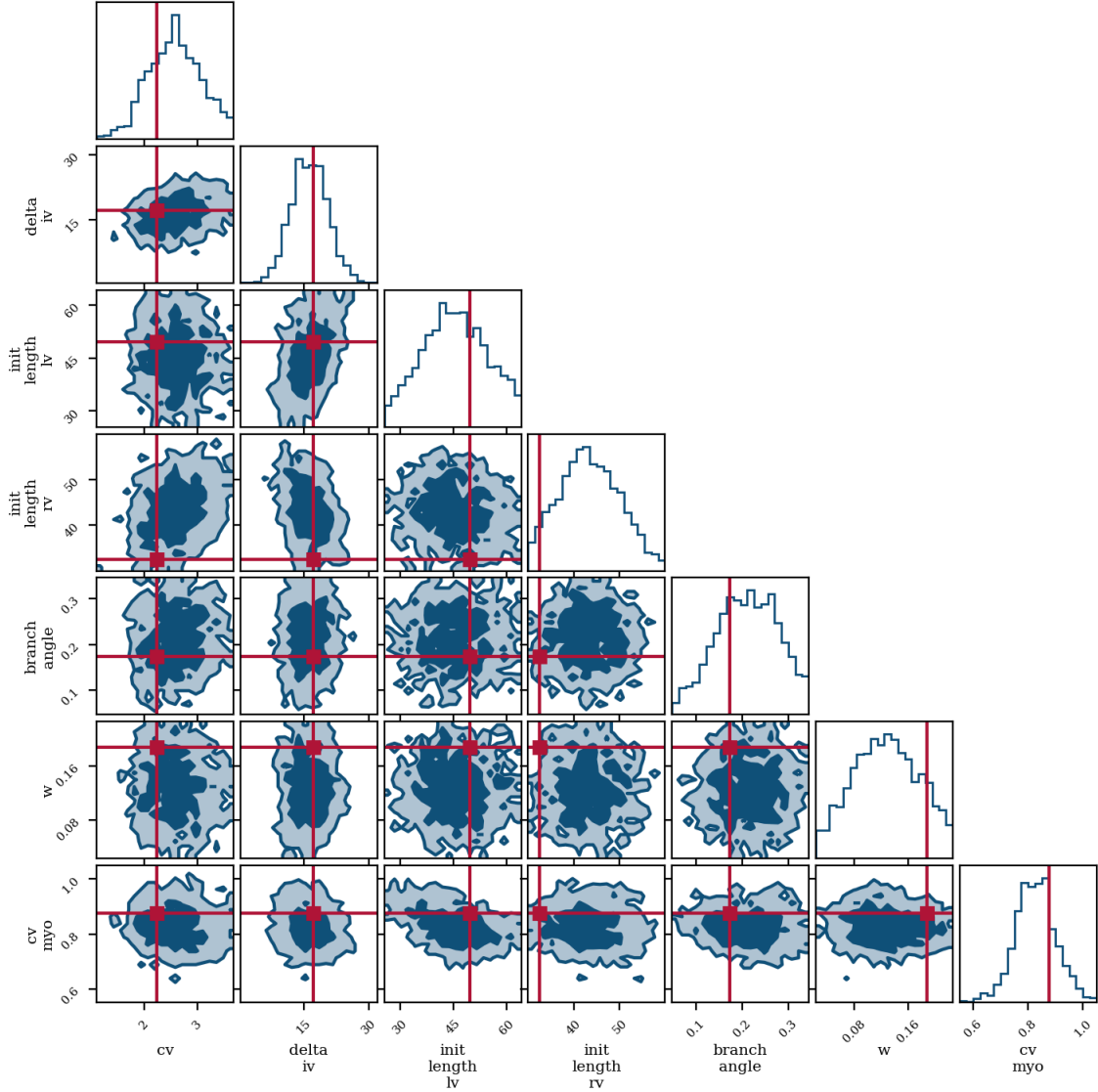


Figure 4: Joint posterior over the seven conduction parameters at a representative operating point (filled contours at 50 and 90 percent; red lines mark the ground truth). Tight marginals (interventricular delay, myocardial velocity) indicate identifiable parameters; broad marginals (branch angle, branch repulsivity, LV extent) the diffuse block.

marginally conservative), with SBC KS median about 0.15. Calibration is necessary but not sufficient for information: it certifies that the diffuse parameters are honestly diffuse, not that the ECG constrains them.

III.C. Budget sensitivity and summary-statistic sufficiency Two readings of the diffuse block are separable, and Figure 6 tests both. Panel (a) recomputes contraction as the training budget grows from 1000 to 4000 draws over three seeds: four parameters (`cv`, `delta_iv`, `init_length_lv`, `init_length_rv`) tighten with budget, a budget-limited rather than information-limited regime at this scale, while the remaining three are flat or widen within seed noise. This is a two-point trend, not a converged curve.

The second reading is that the hand-crafted feature vector is an insufficient summary of the waveform. For any summary statistic S of the data X , the Fisher information obeys $I_S(\theta) \leq I_X(\theta)$, with equality iff S is sufficient

(17,18); summarizing can only destroy information. Panel (b) reports the per-parameter ratio of the feature Cramer-Rao bound to the eight-lead waveform bound, from about 21x to 70x across parameters. The gap is real but roughly common-mode across the spectrum, so it is a candidate summary-statistic insufficiency, testable by a waveform-trained estimator with a learned embedding, not an established ECG information limit.

IV. DISCUSSION

IV.A. What the spectrum is, and is not

The result is a statement about a forward map relative to a stated noise floor, measured on a calibrated posterior. Identifiability here is not a property of a parameter in the abstract: it is the property of *this* forward at *this* floor. The sharpened contribution is exactly that framing. Contraction without a stated floor and without a calibration audit

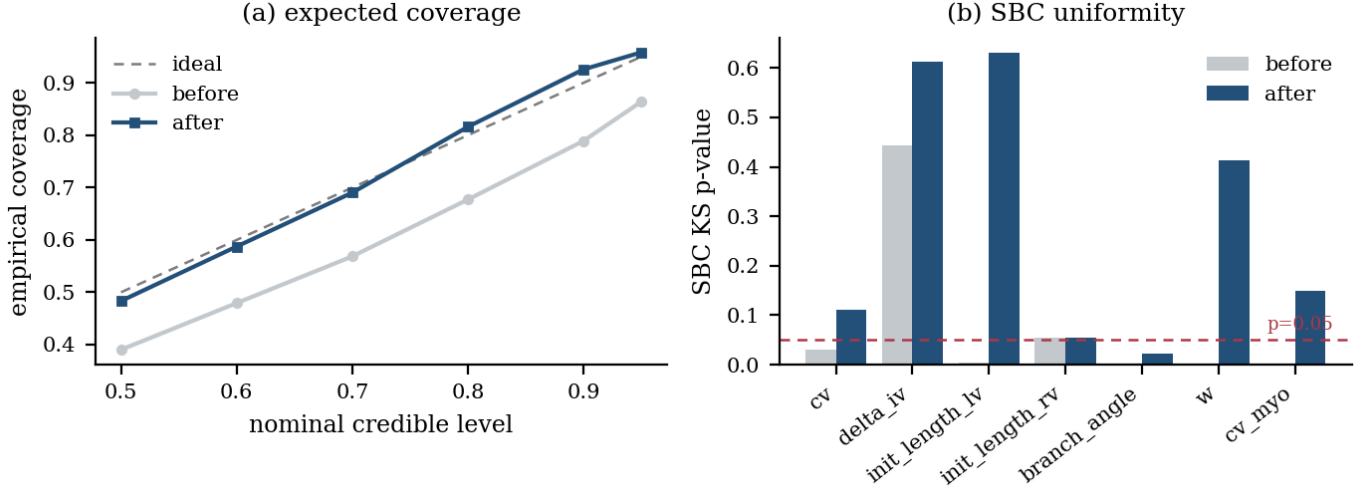


Figure 5: Calibration audit. (a) Expected coverage: before conformal recalibration the empirical coverage falls below nominal (overconfident); after recalibration it tracks the ideal diagonal. (b) SBC uniformity KS p-value per parameter, before and after. The joint TARP ATC moves from -0.057 (overconfident) to $+0.007$ (calibrated).

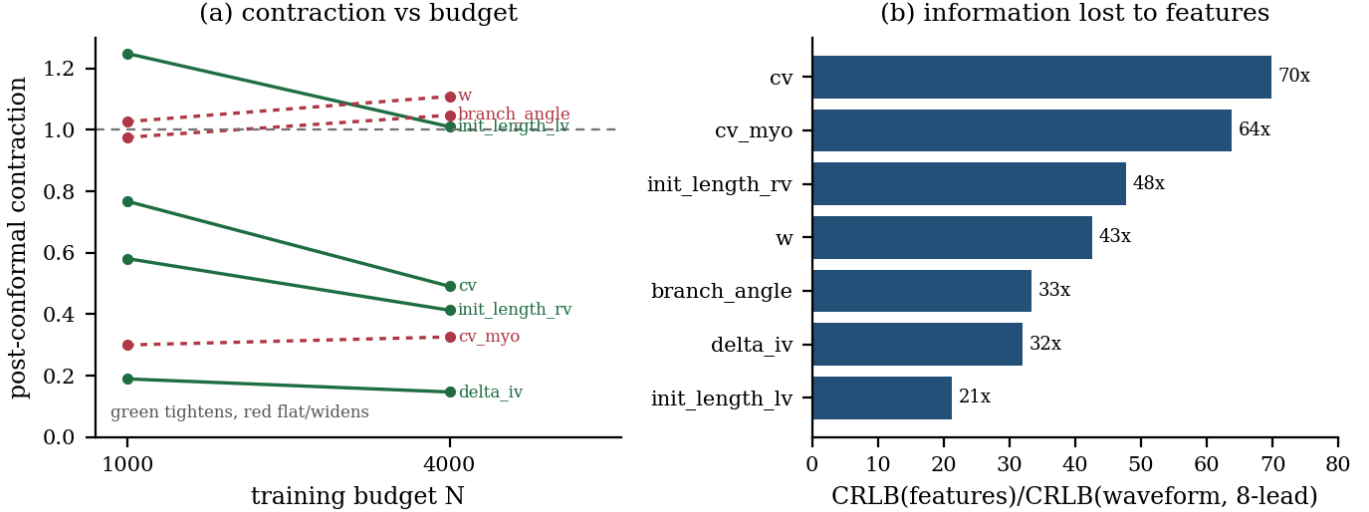


Figure 6: Diffuse-block diagnostics. (a) Contraction versus training budget N (post-conformal median over three seeds): green parameters tighten as budget grows, red parameters are flat or widen within seed noise. (b) Fisher information lost to feature compression, the per-parameter ratio of the feature Cramer-Rao bound to the eight-lead waveform bound; values above one mean the feature summary is less informative than the full waveform.

is uninterpretable, which is why a fit that converges is not evidence of identifiability.

The calibration audit is also where the study corrected itself. A raw contraction spectrum initially looked sharp across all seven parameters; SBC and expected coverage showed the posterior was overconfident, and per-parameter conformal recalibration loosened the diffuse block to at or above the prior width. A posterior correlation between interventricular delay and myocardial velocity (about -0.72) was at first read as a degeneracy and then refuted: a tilted but thin joint is still identifiable, and a correlation is not non-identifiability. An audit further found the spectrum had first been measured in a near-noiseless regime; rescaling the forward to physiological millivolts and re-running

loosened the magnitudes while preserving the ordering and the identifiable-versus-diffuse structure. Each correction narrowed the claim and made it truer.

IV.B. Limitations

No measured ECG appears anywhere in this work; every target is simulator output, so parameter-recovery checks are in the inverse-crime setting and are not fidelity results. The forward is an unbounded homogeneous pseudo-ECG, not a torso model, and its lead-dependent amplitude deficit means amplitude-borne conclusions (most exposed: RV initial extent, which rides a precordial amplitude feature) may not transfer unchanged to a bounded forward. The noise floor is white, whereas real ECG noise is colored; the waveform bound is therefore optimistic in a stated direction of unestablished magnitude. The identifiability

characterization reported here is established on a single reference geometry. Extending it across the publicly available Strocchi virtual cohort is a natural continuation of this work, and preliminary runs on two of those hearts already exercise the full pipeline end to end; we present them as groundwork for a cohort-level study rather than as identifiability claims in their own right.

V. CONCLUSION

We reported a calibrated, amortized characterization of which His-Purkinje and myocardial conduction parameters the 12-lead ECG can determine at fixed anatomy and a stated observation-noise floor. Interventricular delay and myocardial velocity are well constrained, RV initial Purkinje extent and global conduction velocity moderately so, and LV extent, branch angle, and branch repulsivity are formally diffuse, no tighter than the prior. The finding is delivered with a calibration audit that moves the joint posterior from overconfident to calibrated, so the diffuse verdict is an honest information statement rather than an estimator artifact. The methodological point is that identifiability is a property of the forward map at a stated floor, measured on a calibrated posterior, and that reporting a fitted value without that audit cannot distinguish a constrained parameter from an unconstrained one.

V.A. Future work

Three steps, in order of leverage, each trading a stated caveat for a harder test. First, anatomy generalization: repeat the calibrated characterization across a public cohort of geometries to separate what is a property of the forward map from what is a property of one fixed anatomy. Second, a bounded forward: replace the unbounded pseudo-ECG with a torso volume conductor, which changes depolarization amplitude by over an order of magnitude (14), and re-measure the spectrum against that operator. Third, real recordings: move from simulator output to measured 12-lead ECGs, breaking the inverse-crime setting, with a waveform-trained learned embedding as the route past hand-crafted summaries.

V.B. Data and code availability

Code, the trained NPE checkpoint, the simulation sweeps, and the calibration artifacts are released under tag `v0.1.0-submission` at github.com/ricardogr07/ecg-purkinje-npe. The container builds from source per the repository; recorded image digests are in the release notes. An interactive demonstration is available at the project site.

ACKNOWLEDGMENT

The author thanks Professor Francisco Sahli Costabal for the foundational work and guidance that underpin the forward-modeling pipeline used in this study, in particular the fractal Purkinje network generation and eikonal activation on which the identifiability analysis builds (19).

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